

Techniques of integration — Problem Set

Warm-up

1. $\int \sin^2 x \, dx$
2. $\int \frac{e^{1/x}}{x^2} \, dx$
3. $\int \frac{1}{\sqrt{1-x^2}} \, dx$
4. $\int \frac{1}{\sqrt{9-x^2}} \, dx$
5. $\int \frac{1}{1+x^2} \, dx$
6. $\int x(x+1)^5 \, dx$
7. $\int \frac{\ln x}{x} \, dx$
8. $\int e^{\cos x} \sin x \, dx$
9. $\int x e^x \, dx$
10. $\int \ln x \, dx$

Skill-building

1. Evaluate $\int_0^{\pi/4} \cos^2 2x \, dx$.
2. Use the substitution $u = 2x + 1$ to find $\int x(2x+1)^4 \, dx$.
3. Find the exact value of $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} \, dx$.
4. Find $\int \frac{1}{\sqrt{16-9x^2}} \, dx$.
5. Use the substitution $u = e^x$ to evaluate $\int \frac{e^x}{1+e^{2x}} \, dx$.
6. Evaluate $\int_0^{\pi/2} x \sin x \, dx$ using integration by parts.

7. Show that $\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx$.
8. Use the substitution $u = \ln x$ to evaluate $\int_1^e \frac{(\ln x)^2}{x} dx$.
9. Find $\int \tan^{-1} x dx$ using integration by parts.
10. Use the substitution $u = 4 - x^2$ to find $\int x \sqrt{4 - x^2} dx$.

Easier Exam Questions

1. (Caringbah 2021 Q11a) Evaluate **3**

$$\int_0^{\frac{\pi}{6}} \sin^2 2x dx$$

2. (Caringbah 2021 Q11b) Using the substitution $u = \frac{1}{x}$, find the exact value of **3**

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$$

3. (Sydney Girls 2022 Q11b) Find the value of the following integral, giving your answer in terms of π . **2**

$$\int_0^{\frac{1}{3}} \frac{1}{\sqrt{4 - 9x^2}} dx$$

4. (Caringbah 2022 Q11c) Use the substitution $u = x + 1$ to evaluate **3**

$$\int_1^3 \frac{x + 2}{(x + 1)^2} dx$$

Express your answer in simplest exact form.

5. (Cherrybrook 2022 Q11d) Use the substitution $u = 2x - 1$ to evaluate $\int_{\frac{1}{2}}^2 x \sqrt[3]{2x - 1} dx$. **3**

Write your answer correct to one decimal place.

6. (Gosford 2025 Q11d) Use the substitution $x = 2 \cos u$ to evaluate $\int_0^1 \frac{x^2}{\sqrt{4 - x^2}} dx$. **3**

7. (Baulkham Hills 2025 Q11c) Find $\int \frac{dx}{\sqrt{25 - 4x^2}}$. **2**

8. (Sydney Girls 2020 Q11b) Find: **2**

$$\int \frac{dx}{\sqrt{12 - 4x^2}}$$

9. (Sydney Tech 2025 Q11b) Find $\int_0^{\ln 2} \frac{e^{2x}}{1 + e^{4x}} dx$ by using the substitution $u = e^{2x}$, correct to 2 decimal places. **4**

10. (Ascham 2020 Q13b) Use the substitution $u = x \ln x$ to evaluate $\int_1^2 \frac{\ln x^2 + 2}{(x \ln x - 1)^2} dx$. **3**

11. (Abbotsleigh 2023 Q11b) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$. **2**

12. (Girraween 2023 Q11b) Use the substitution $u = 6 - x$ to find the exact value of **3**

$$\int_1^6 x\sqrt{6-x} dx$$

13. (Girraween 2023 Q12a) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$ **3**

14. (Killara 2021 Q11d) Evaluate the following integral using the substitution $u = 1 - x^4$. **2**

$$\int_0^{\frac{1}{\sqrt{2}}} \frac{2x^3}{\sqrt{1-x^4}} dx$$

15. (Sydney Grammar 2023 Q11d) (d) Evaluate $\int \cos^2 2x dx$. **2**

16. (Sydney Grammar 2023 Q12a) Calculate $\int_0^1 x(1-x)^7 dx$ using the substitution $u = 1 - x$. **3**

17. (Cherrybrook Tech 2021 Q12c)

Evaluate $\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} \cos^2 2\theta \sin 2\theta d\theta$. Use the substitution $u = \cos 2\theta$. **3**

18. (St George Girls 2020 Q13a) Find $\int_0^{\ln 2} \frac{e^{2x}}{1+e^{4x}} dx$ by using the substitution $u = e^{2x}$, to two decimal places. **3**

19. (Girraween 2022 Q11c) Using the substitution $u = \frac{1}{x}$, find the exact value of **3**

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$$

20. (Ashcham 2024 Q11a) $\int \frac{1}{\sqrt{4-9x^2}} dx$. **2**

21. (Ashcham 2024 Q11d) Find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos^2 x dx$. **3**

22. (Ashcham 2024 Q12c) Use the substitution $u = \sqrt{2-x}$ to find the exact value of **3**

$$\int_0^1 \frac{x}{\sqrt{2-x}} dx.$$

Harder Exam Questions

1. (Riverview 2020 Q11b) Use the substitution $u = 9 - x^2$, to find the primitive function of **2**

$$\frac{x dx}{\sqrt{9-x^2}}$$

2. (Sydney Girls 2020 Q11c) Use the substitution $x = \cos \theta$ (where $0 < \theta < \frac{\pi}{2}$) to determine:

$$\int \sqrt{\frac{x^2}{1-x^2}} dx.$$

3

3. (Penrith 2024 Q12c) Find $\int \frac{\ln x}{x} dx$, using the substitution $u = \ln x$. **2**

4. (Gosford 2020 Q11a) Use the substitution $u = \cos x + 1$, find the exact value of the following integral: **2**

$$\int_0^{\frac{\pi}{2}} e^{\cos x + 1} \sin x \, dx$$

5. (Caringbah 2022 Q12d)

(i) Use the substitution $x = \tan^2 \theta$ to show that **2**

$$\int_0^1 \frac{\sqrt{x}}{(1+x)^2} \, dx = \int_0^{\frac{\pi}{4}} 2 \sin^2 \theta \, d\theta$$

(ii) Hence, or otherwise, find in simplest exact form the value of **2**

$$\int_0^1 \frac{\sqrt{x}}{(1+x)^2} \, dx$$

6. (Girraween 2022 Q11d) Evaluate $\int_0^{\sqrt{5}} \frac{2x^3}{\sqrt{x^2+4}} \, dx$ using the substitution $u = x^2 + 4$ **4**